Assignment 1: Text Analysis

36118: Applied Natural Language Processing

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Table Of Contents

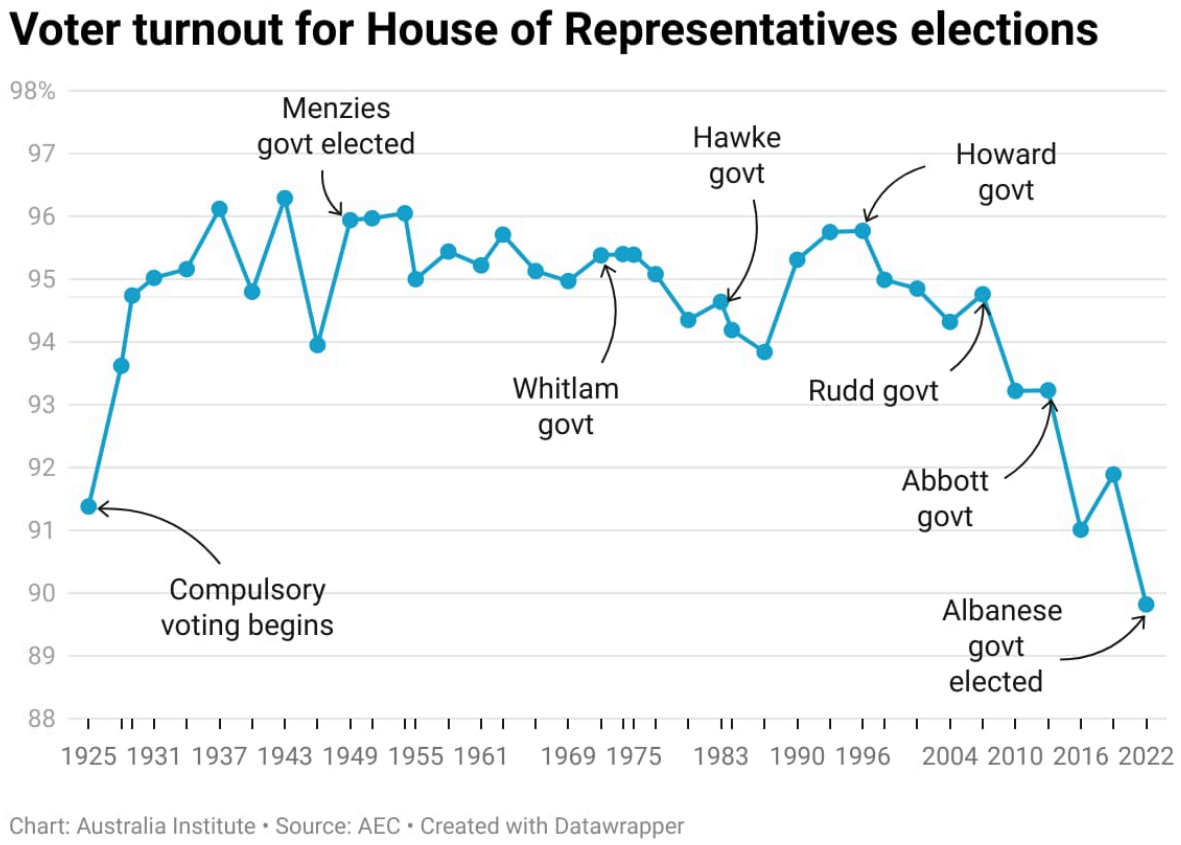
Section 1: Introduction & Objectives---------- 3
1.1 Introduction---------- 3
1.2 Aims and Objectives of the Project:---------- 5
1.3 Research Questions:----------6
Section 2: Data Ingestion and Insights---------- 7
2.1 Dataset Overview----------7
2.2 Data Ingestion:----------7
2.3 Data Insights:---------- 7
2.4 Optical Character Recognition (OCR):---------- 8
Section 3: Cleaning and Pre-processing---------- 9
3.1 Basic Text Cleaning:----------9
3.2 Sanity Checks:----------9
Section 4: Tokenization, Stop Words Removal, and Lemmatization---------- 10
4.1 Tokenisation and Stopword Removal :----------10
4.2 Lemmatization:----------10
Section 5: Word Frequencies---------- 11
5.1 Most Frequent Words:----------11
5.2 Word Cloud:----------11
5.3 Key Word Search:----------12
Section 6: N-Grams and TF-IDF----------13
6.1 Bi-grams:---------- 13
6.2 Tri-grams:----------13
6.3 TF-IDF:----------14
Section 7: Topic Modelling---------- 15
Section 8: Named Entity Recognition (NER) and Sentiment Analysis----------16
Section 9: BERTopic Modelling:----------18
Section 10: Inference and Recommendations:---------- 19
10.1 Inference:---------- 19
10.2: Recommendations:----------20

Section 1: Introduction & Objectives

1.1 Introduction

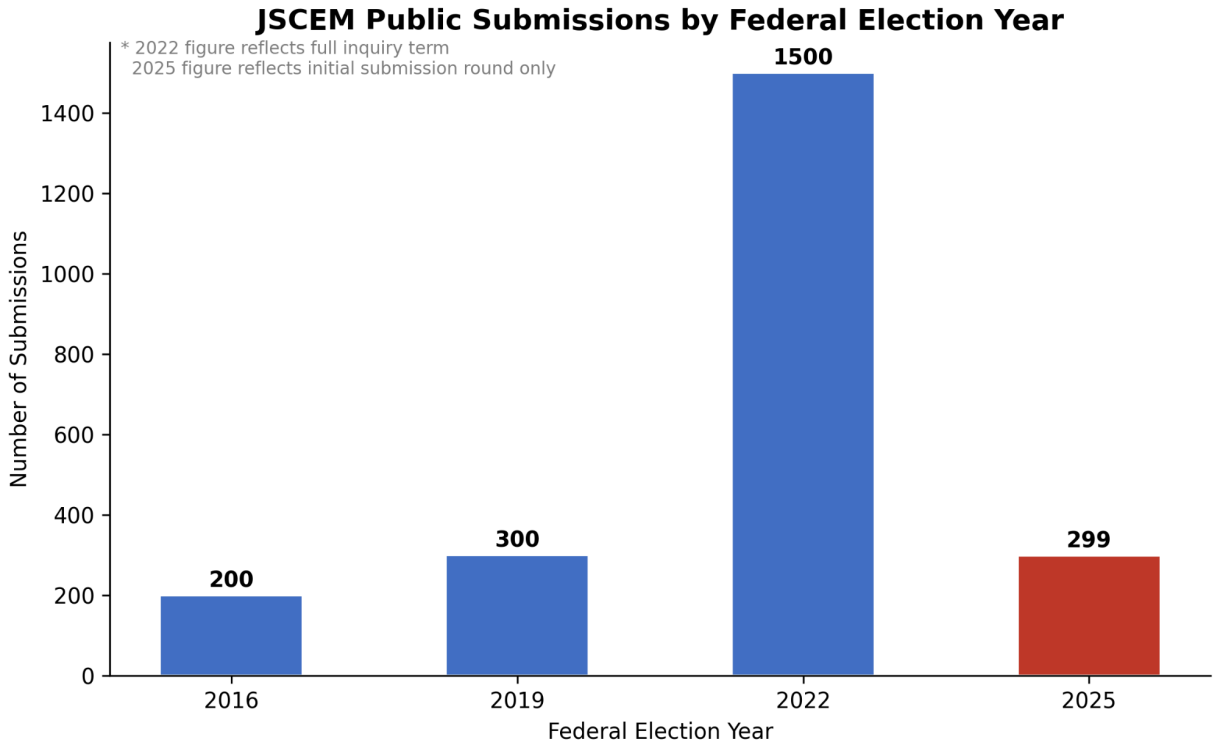
(The need for analysing voter turnout, voter sentiment, and post-election public opinion has never been greater)

Australia's federal elections are among the most regulated and scrutinised democratic processes in the world. With compulsory voting, preferential balloting, and an independent electoral commission, the Australian system is often cited internationally as a model of electoral integrity. Yet every election cycle surfaces new tensions about how votes are cast, how candidates campaign, and whether the rules governing elections are keeping pace with a rapidly changing social and technological landscape.



Adding to the urgency of this inquiry is a long term decline in voter participation that shows no signs of reversing. Despite compulsory voting, formal voter turnout for House of Representatives elections has fallen from a peak of over 96% in the mid-20th century to just **89.8% in 2022**, the lowest recorded since compulsory voting was introduced in 1925. This steady erosion, as shown in the chart above, sourced from the Australia Institute using AEC data, suggests that even the legal obligation to vote is no longer sufficient to maintain full democratic participation. When fewer Australians are engaging at the ballot box, the question of *why* and what the electoral system can do about it becomes exactly the kind of question a public inquiry should be answering.

This, coupled with the Survey evidence, shows most Australians still say democracy is the best form of government, but there is persistent concern and distrust about how politics actually works. That combination of broad support for democracy plus scepticism about institutions is exactly the environment where people are motivated to write submissions to inquiries like JSCEM to "fix the system" rather than abandon it. Following every federal election since 1984, the Joint Standing Committee on Electoral Matters (JSCEM) has invited the Australian public to formally submit their views on how the election was conducted.



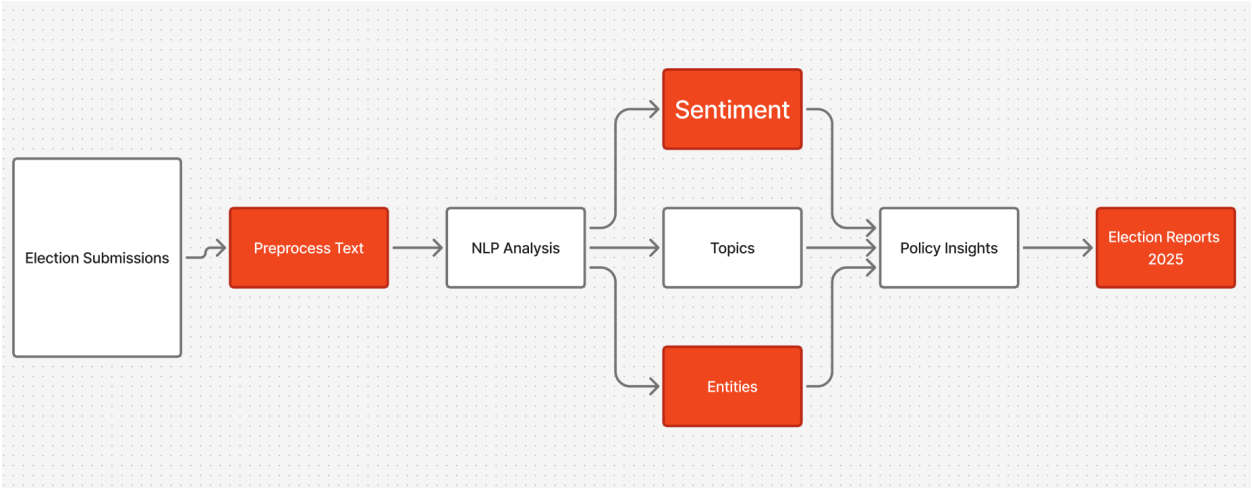
Australians have shown significant participation the 2022 spike to over 1,500 submissions reflects a turning point. The committee received more than 1,500 submissions and held 11 public hearings, making 36 recommendations on issues ranging from donation disclosure to truth in political advertising. This scale of public engagement was unprecedented for a JSCEM inquiry and signals a broader shift in how actively Australians are engaging with electoral governance.

With public engagement at historic highs and trust in institutions at historic lows, analysing what Australians are actually saying in these submissions at scale, objectively, and without bias has never been more important.

1.2 Aims and Objectives of the Project:

(How we solve the analysis problem with Natural Language Processing)

NLP provides a practical way to work through a large volume of text and find what is actually happening within it by identifying recurring themes, understanding how people feel, and recognising who or what they are talking about. Instead of manually reading hundreds of documents, computational techniques allow patterns, topics, and sentiments to emerge from the data in a structured and scalable way.



(NLP Pipeline followed. Source: Made with Figma)

This project applies these techniques to the 299 public submissions lodged with the 2025 Australian Federal Election inquiry. The dataset captures a wide range of perspectives, from individual citizens describing their experiences at polling booths to major organisations and institutions presenting detailed policy recommendations and electoral reform proposals.

By applying natural language processing methods such as text preprocessing, topic identification, and sentiment analysis, the project aims to make sense of this diverse body of submissions. The goal is not only to summarise what people are saying, but also to reveal broader patterns in public concerns, priorities, and discussions surrounding the electoral process. In doing so, NLP helps transform a large and complex collection of written submissions into insights that can highlight the issues most frequently raised, the tone of public engagement, and the key actors or institutions mentioned throughout the inquiry.

1.3 Research Questions:

Although there are several ways to analyze this project, my project aims to answer these 3 research questions:

RQ1: What are the dominant themes and concerns raised by the Australian public about the conduct of the 2025 federal election?

RQ2: Is there evidence of growing public concern about disinformation and online political advertising in Australian elections?

RQ3: Are all voices represented equally in this inquiry, or do institutional submitters dominate the conversation over individual citizens?

Section 2: Data Ingestion and Insights

2.1 Dataset Overview

The dataset used in this project consists of 299 public submissions lodged with the Joint Standing Committee on Electoral Matters as part of the 2025 Australian Federal Election inquiry. These submissions were collected through a formal parliamentary consultation process open to anyone which means the corpus captures a genuinely diverse range of voices, from a retired citizen writing a page about their experience at the polling booth, to the Australian Electoral Commission submitting a 10,000 word policy document, to religious organisations, advocacy groups, and political parties all making their case in the same forum.

Each submission was provided as a PDF file, named in a consistent format that identifies the submitter or flags where a name was withheld. The submissions vary enormously in length and structure. Some are brief personal statements of a few hundred words. Others are formal reports with executive summaries, appendices, and numbered recommendations running to dozens of pages. From a technical standpoint, the data is moderately sized, with 302,712 words across a vocabulary of 14,114 unique terms after cleaning, averaging 1,104 words per submission.

2.2 Data Ingestion:

1. The data ingestion process began with locating all 299 PDF submissions stored in a local project directory. Using Python's glob library, the files were automatically indexed by scanning the folder and collecting all PDF paths into a list.
2. Text extraction was handled by pdfplumber, which reads each PDF page by page and pulls out the raw text. For each submission, page text was concatenated into a single continuous string and stored alongside its filename in a pandas DataFrame.
3. Because running pdfplumber across 299 files takes several minutes, the resulting DataFrame was immediately saved to a pickle file after the first extraction. Every subsequent session loads directly from that pickle.
4. The output of this stage was a clean two-column DataFrame filename and text containing all 299 submissions, which served as the input for every subsequent stage of the pipeline.

2.3 Data Insights:

An initial word count inspection across all 299 submissions immediately revealed how uneven the corpus is. The mean submission length was 1,979 words, but the standard deviation of 3,795 tells the more important story, a small number of very long institutional documents are pulling that average up significantly, while the median of just 831 words reflects what most individual submitters actually wrote. The longest submission ran to 36,258 words. The shortest returned just 8. At the lower end, 21 submissions came back with fewer than 50 words. Inspecting these manually confirmed they were not genuine short submissions they contained only repeated

header text along the lines of *"Inquiry into the 2025 federal election Submission 119"* with no actual content beneath it. These are almost certainly scanned image based PDFs where pdfplumber could read the document header embedded in the file metadata, but could not extract the page content itself, because the pages are images rather than selectable text.This distinction matters because a submission returning 8 words is not a citizen who had nothing to say it is a citizen whose submission we have not yet successfully read. These 21 files were flagged for OCR processing in the next stage rather than being dropped from the analysis, ensuring no submission was excluded simply because of a technical extraction failure.

2.4 Optical Character Recognition (OCR):

After identifying documents with very low word counts, a threshold of 50 words was used to flag submissions that likely contained little or no extracted text. These files were considered potential candidates for Optical Character Recognition (OCR), as they might contain scanned images rather than selectable text. The filenames of these submissions were retrieved and their full file paths were constructed to verify that the documents existed in the source directory. OCR tools were then configured by specifying the paths to Tesseract OCR and the Poppler binaries required for PDF image conversion. This step ensured that PDFs with minimal extracted text could be reprocessed using OCR if necessary, helping recover text that standard PDF extraction methods may have missed and improving the overall completeness of the dataset. The OCR pipeline successfully recovered text from scanned submissions. The word counts for previously empty files increased substantially.

```
#Inspecting the documents containing less than 50 words
df[df['word_count'] < 50][['filename', 'word_count']]
```

[8] ✓ 0.0s

	filename	word_count
14	Sub015 - Name withheld.pdf	8
15	Sub016 - Name withheld.pdf	8
32	Sub033 - Name withheld.pdf	8
51	Sub052 - John Ferguson.pdf	8
56	Sub057 - Tony Robinson.pdf	8
69	Sub072 - Lynn Lawson.pdf	8
74	Sub077 - Name withheld.pdf	8
76	Sub079 - Jane Johnson.pdf	8
116	Sub119 - Name Withheld.pdf	40
147	Sub148 - Advance Australia.pdf	32
151	Sub152 - National Party of Australia.pdf	16
177	Sub179 - David Robertson.pdf	8
182	Sub184 - Robert Kimpton.pdf	20
185	Sub189 - Nick Toovey.pdf	8
241	Sub249 - Name Withheld.pdf	16
253	Sub263 - Troy Harper.pdf	8
260	Sub270 - Michael Butterworth.pdf	24
261	Sub271 - Geoff Grimes.pdf	16
262	Sub272 - Name Withheld.pdf	39
279	Sub292- Name withheld.pdf	16
290	SupSub093.1 - Name withheld.pdf	22

```
#Checking to see if texts were recovered from the PDF
print(f"df has {len(df)} rows")
df[df['word_count'] < 50][['filename', 'word_count']]
```

✓ 0.0s

df has 299 rows

	filename	word_count
262	Sub272 - Name Withheld.pdf	33

```
Sub015 - Name withheld.pdf: 290 words recovered
Sub016 - Name withheld.pdf: 410 words recovered
Sub033 - Name withheld.pdf: 420 words recovered
Sub052 - John Ferguson.pdf: 319 words recovered
Sub057 - Tony Robinson.pdf: 390 words recovered
Sub072 - Lynn Lawson.pdf: 96 words recovered
Sub077 - Name withheld.pdf: 165 words recovered
Sub079 - Jane Johnson.pdf: 348 words recovered
Sub119 - Name Withheld.pdf: 1650 words recovered
Sub148 - Advance Australia.pdf: 1631 words recovered
Sub152 - National Party of Australia.pdf: 632 words recovered
Sub179 - David Robertson.pdf: 263 words recovered
Sub184 - Robert Kimpton.pdf: 338 words recovered
Sub189 - Nick Toovey.pdf: 422 words recovered
Sub249 - Name Withheld.pdf: 751 words recovered
Sub263 - Troy Harper.pdf: 335 words recovered
Sub270 - Michael Butterworth.pdf: 561 words recovered
Sub271 - Geoff Grimes.pdf: 143 words recovered
Sub272 - Name Withheld.pdf: 33 words recovered
Sub292- Name withheld.pdf: 695 words recovered
SupSub093.1 - Name withheld.pdf: 358 words recovered
```

Section 3: Cleaning and Pre-processing

3.1 Basic Text Cleaning:

Before any analysis could begin, the raw extracted text needed to be cleaned. The cleaning process worked through the text in a logical sequence. Everything was lowercased first so that Election and election were treated as the same word. Non-breaking spaces and formatting characters were normalised. The phrase "Inquiry into the 2025 federal election" and submission number headers were stripped out as they appeared across almost every document but carried no analytical value. URLs, email addresses, and page number references were removed for the same reason. What remained was then filtered down to alphabetic characters and spaces only, removing numbers and punctuation that would otherwise clutter word frequency counts. Multiple consecutive spaces were collapsed and any leading or trailing whitespace was trimmed.The cleaned text was saved into a new clean_text column rather than overwriting the original, so the raw extracted text remains available throughout the project.

```
df[['clean_text', 'filename']].head(15)
```

	clean_text	filename
0	time to revisit territory representation intro...	Sub001 - Bob McMullan.pdf
1	possible solutions more money or fewer core re...	Sub002 - Dr Sheppard.pdf
2	submission to the september submission to the ...	Sub003 - Grattan Institute.pdf
3	aec submission joint standing committee on ele...	Sub004 - Australian Electoral Commission.pdf
4	australian democracy in submission to the inqu...	Sub005 - The Australia Institute.pdf
5	submission to inquiry into election committee ...	Sub006 - Andrew Murray.pdf
6	written submission the joint standing committe...	Sub007 - Genevieve Grant.pdf
7	ten years ago epping was the first suburb in s...	Sub008 - Richard Ure.pdf
8	election submission by barry searle sept it s ...	Sub009 - Barry Searle.pdf
9	my submission regarding the federal election i...	Sub010 - Name withheld.pdf
10	committee secretary bruce waddell joint standi...	Sub011 - Bruce Waddell.pdf
11		Sub012 - Dr Matthew Jennings.pdf
12	submission to the joint standing committee on ...	Sub013 - Lex Stewart.pdf
13	from monica sammon to committee em reps subjec...	Sub014 - Monica Sammon.pdf
14	from r to committee em reps subject name withh...	Sub015 - Name withheld.pdf

Generate + Code + Markdown

3.2 Sanity Checks:

After cleaning, a final quality check was applied to catch any submissions that still contained too little content to be worth analysing. A word count on the cleaned text was calculated for each document, and anything under 20 words was flagged as non-substantive. These were typically submissions where OCR had partially failed or where the original document contained nothing beyond a cover page. Rather than dropping these rows from the DataFrame entirely, a boolean is_substantive column was added so the full dataset remains intact and the filtering decision is transparent. A separate analysis_df was then created containing only the 274 substantive submissions, which serves as the working corpus for all NLP analysis that follows.

Section 4: Tokenization, Stop Words Removal, and Lemmatization

4.1 Tokenisation and Stopword Removal :

Once the text was clean, each submission was tokenised into individual words using NLTK's word_tokenize. Non alphabetic tokens, numbers, punctuation, and characters were discarded, and standard English stopwords like the, is, and of were removed since they appear everywhere but tell us nothing about what a submission is actually about. The resulting tokens were stored in a new tokens_no_lemma column, keeping both the cleaned text and its tokenised form available for the frequency analysis, topic modelling, and n-gram extraction that follows.

	filename	clean_text	tokens_no_lemma
0	Sub001 - Bob McMullan.pdf	time to revisit territory representation intro...	[time, revisit, territory, representation, int...
1	Sub002 - Dr Sheppard.pdf	possible solutions more money or fewer core re...	[possible, solutions, money, fewer, core, resp...
2	Sub003 - Grattan Institute.pdf	submission to the september submission to the ...	[submission, september, submission, aruna, sat...
3	Sub004 - Australian Electoral Commission.pdf	aec submission joint standing committee on ele...	[aec, submission, joint, standing, committee, ...
4	Sub005 - The Australia Institute.pdf	australian democracy in submission to the inqu...	[australian, democracy, submission, inquiry, f...

(Initial Row check to show the result of Stop Word Removal and Tokenization)

4.2 Lemmatization:

In this step, lemmatization was applied to the tokenized text to reduce words to their base or dictionary form, helping group different grammatical variations of the same word together. Using NLTK's WordNetLemmatizer, each token was first assigned a part-of-speech (POS) tag through the pos_tag function. These POS tags were then mapped to the corresponding WordNet categories (noun, verb, adjective, or adverb) to ensure that the lemmatizer used the correct grammatical context when reducing words. For example, words such as voting, voted, and votes could be normalized to the base form vote. The resulting lemmatized tokens were stored in a new column (tokens_lemmatized) while preserving the original tokenized version.

```
BEFORE LEMMATIZATION:

['time', 'revisit', 'territory', 'representation', 'introduction', 'act', 'senate', 'representation', 'anomalous', 'election', 'arrangements', 'years', 'time',

AFTER LEMMATIZATION:

['time', 'revisit', 'territory', 'representation', 'introduction', 'act', 'senate', 'representation', 'anomalous', 'election', 'arrangement', 'year', 'time',
```

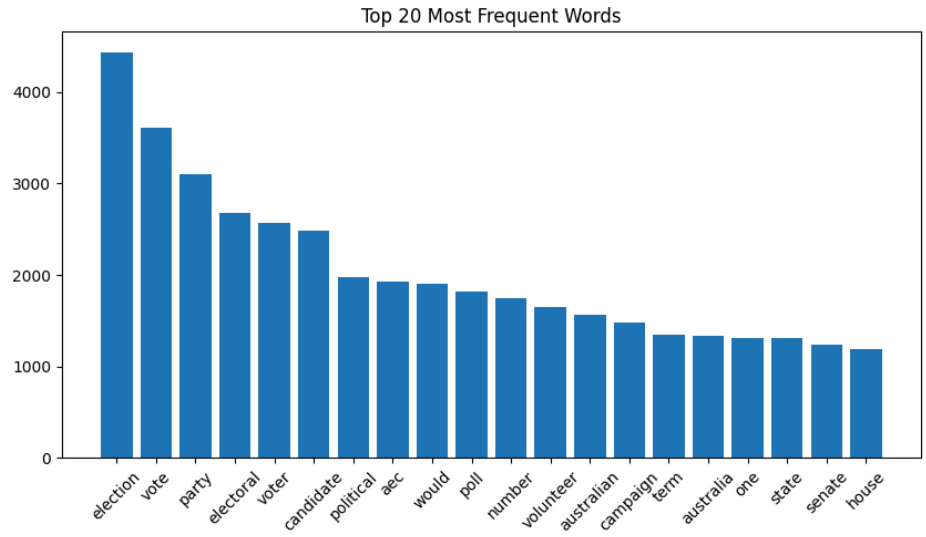
(Results of Lemmatization)

Section 5: Word Frequencies

5.1 Most Frequent Words:

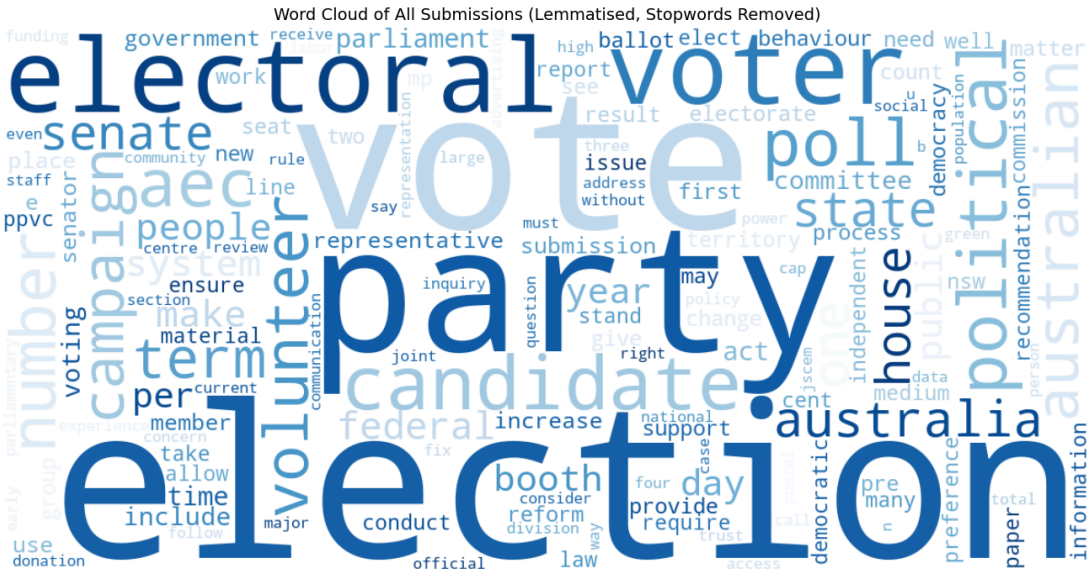
With the text cleaned, tokenised, and lemmatised, the first analytical step was a straightforward word frequency count across the entire corpus. All tokens from every submission were pooled together and counted using Python's Counter, giving a corpus-level view of which terms appear most frequently and how large and diverse the vocabulary is overall. The top 20 most frequent terms were then visualised as a bar chart.

Insights: The top 20 most frequent words tell a clear story. Election and vote dominate by a significant margin, which is expected given the subject matter, but what is more revealing is what comes immediately after party, electoral, voter, candidate, political, and aec all appear over 1,800 times each. This suggests submissions are not just describing the election in general terms but are actively engaging with the mechanics of how it was run, who stood, who voted, how they voted, and what the AEC did or did not do well. The presence of volunteer, poll, booth, and campaign in the top 20 further confirms that polling day operations are a central preoccupation across the corpus a finding that will be reinforced consistently throughout the topic modelling and n-gram analysis that follows.



5.2 Word Cloud:

To complement the bar chart, a word cloud was generated across all lemmatised tokens from the full corpus after stopword removal. Word size is proportional to frequency, making it easy to spot dominant terms at a glance while also picking up mid-frequency words that a ranked bar chart would push out of view. It is a deliberately exploratory visual, not a precise measurement, but a useful way to get a feel for the overall language of the corpus before moving into more structured analysis.



Insights: The word cloud reinforces the bar chart findings while adding useful nuance. Volunteer, booth, poll, and ppvc sit prominently alongside institutional terms like aec, Senate, and Parliament, revealing a corpus simultaneously concerned with the ground level experience of election day and the broader structural questions about how Australian democracy is organised. The visibility of reform, trust, and behaviour signals that submitters are not simply describing what happened, they are making judgements about it.

5.3 Key Word Search:

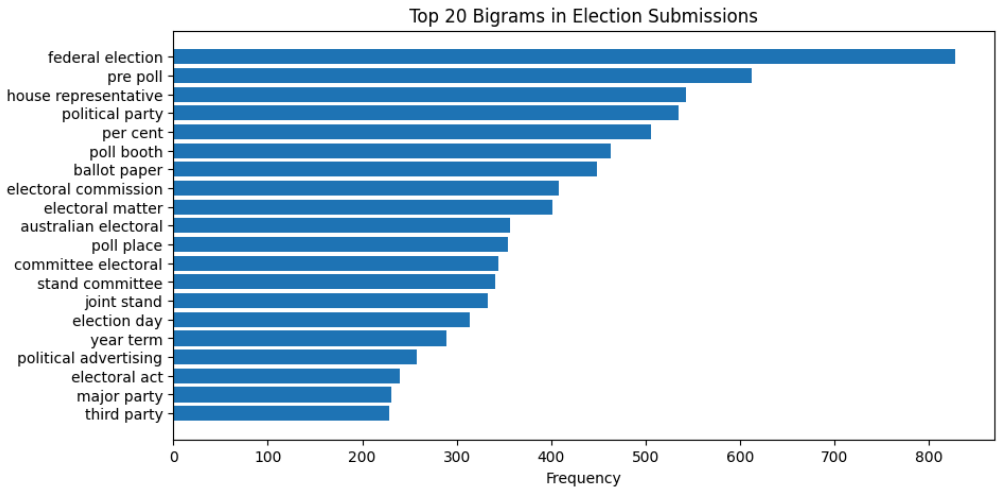
To move beyond general frequency counts, a targeted keyword search was conducted across all submissions to measure how widely specific policy-relevant terms appear in the corpus. Rather than counting total occurrences, this analysis measures how many individual submissions contain each term, giving a truer picture of how broadly a concern is shared across the public rather than how loudly a single submitter raises it.

Insights: The results are revealing. Reform appears in 115 submissions more than any other keyword and nearly half the entire working corpus, confirming it as the single most cross cutting concern raised by the public. Vote (246) is expected given the subject matter, but the breadth of postal (77), integrity(79), and trust (75) is more significant, these are not niche concerns raised by a handful of submitters, they are widespread. By contrast, fraud appears in just 6 submissions, which is an important finding in itself, despite frequent political rhetoric about electoral fraud, the public engaging with this inquiry is far more concerned about systemic reform and institutional trust than about specific allegations of misconduct.

Section 6: N-Grams and TF-IDF

6.1 Bi-grams:

While single word frequency analysis identifies the most common terms, bigram analysis identifies pairs of consecutive words that reveal more meaningful phrases and concepts. A bigram analysis was conducted across all lemmatised tokens to surface the most frequently occurring two-word combinations, giving a clearer picture of the specific issues and processes submitters are discussing rather than isolated keywords.



Insights: Pre poll coming in second overall, ahead of house representative and political party, which confirms pre-poll voting as the single most discussed practical concern in the corpus, with poll booth, ballot paper, and election day all reinforcing that the public is focused on the physical mechanics of how votes are cast rather than abstract electoral theory. The presence of political advertising and electoral acts signals a subset of submitters engaging at a formal legislative level, pointing to genuine demand for regulatory change rather than just general dissatisfaction. Notably, third party and major party both appearing in the lower half hints at growing concern about the role of minor parties and independent candidates in shaping electoral outcomes, a theme that became particularly prominent in the 2022 and 2025 elections with the rise of the teal independents. This directly addresses **RQ1,** the bigram analysis confirms that pre-poll voting operations and electoral mechanics are the dominant practical concerns raised across the corpus.

6.2 Tri-grams:

Trigram analysis extends the n-gram approach to three word sequences, surfacing specific phrases and policy references that bigrams cannot fully capture. These longer combinations tend to represent named entities, legislative references, and recurring procedural concepts that reveal how submitters are formally framing their concerns.

Insights: The top four trigrams are all variations of the same institutional reference the committee and the AEC, confirming that submitters are acutely aware they are writing to a formal parliamentary body. More significant is truth political advertising appearing 143 times, which can only be this frequent because a concentrated group of submitters is specifically and deliberately raising it as a policy demand strong evidence supporting **RQ2** that concern about misleading political content represents a distinct and organised strand of public opinion. The cluster of PPVC related trigrams all appearing exactly 81 times suggests they originate from a single submission or a small group of near-identical ones, pointing back to **RQ1** and the dominance of pre-poll voting as the corpus's central practical concern.

6.3 TF-IDF:

TF-IDF (Term Frequency Inverse Document Frequency) goes beyond raw word counts by measuring how distinctive a term is to individual submissions relative to the entire corpus. Words that appear everywhere score low regardless of frequency, while terms that are concentrated in specific submissions score high, which makes it a better tool for identifying what each submission is uniquely about rather than what the corpus shares in common.

Insights: The TF-IDF results tell a more nuanced story than the raw frequency counts. Volunteer scores highest by a clear margin, meaning it is not just frequently mentioned but is the most distinctively discussed term at the individual submission level, suggesting a concentrated group of submitters with specific and detailed concerns about polling booth volunteering. Party, voter, poll, and candidate follow closely, confirming these as the core substantive themes of the inquiry rather than just common background vocabulary. The appearance of behaviour in the top 20 shows thatit is not a high-frequency word in absolute terms, but its TF IDF score indicates it appears in a focused cluster of submissions, referring to the conduct of candidates, party volunteers, or voters at polling booths, which directly connects to **RQ1**. The bigram pre poll also making the top 20 further reinforces that pre-poll voting is not just frequently mentioned but is a distinctively important topic for specific submitters engaging with it in depth.

	term	mean_tfidf
0	volunteer	0.095450
1	party	0.071097
2	voter	0.070445
3	poll	0.070187
4	candidate	0.068229
5	booth	0.059165
6	electoral	0.058711
7	aec	0.050074
8	campaign	0.047276
9	political	0.046384
10	number	0.040603
11	day	0.037102
12	term	0.036384
13	pre	0.036184
14	people	0.035579
15	committee	0.034963
16	australian	0.034542
17	pre poll	0.034430
18	behaviour	0.033029
19	australia	0.031917

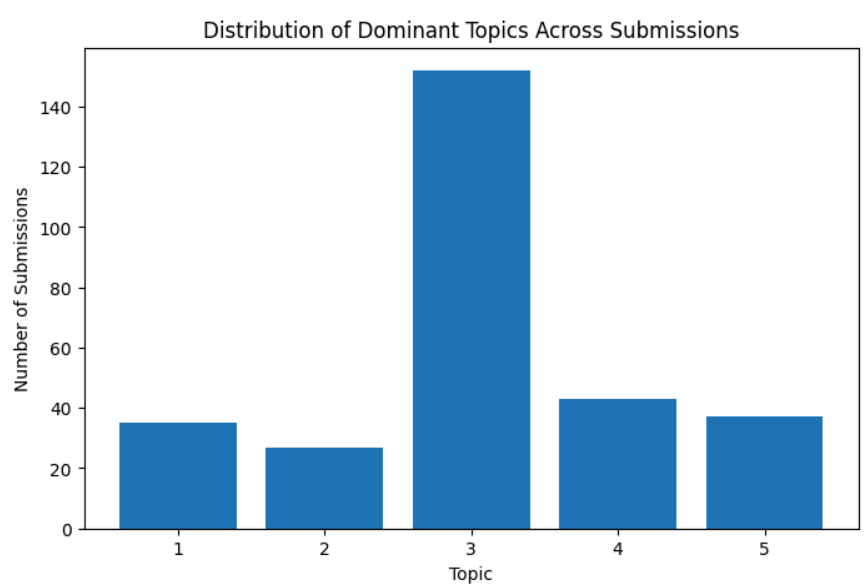
(Results of TF-IDF)

Section 7: Topic Modelling

Latent Dirichlet Allocation (LDA) is a probabilistic topic modelling technique that identifies latent themes within a corpus by assuming each document is a mixture of topics and each topic is a distribution over words. Rather than reading submissions one by one, LDA groups them automatically based on shared vocabulary patterns, surfacing the underlying themes present across the entire corpus. Five topics were selected after testing multiple configurations, as this number produced the most interpretable and distinct groupings for this dataset.

```
Topic 1: vote, senate, house, number, would, state, senator, per, system, one
Topic 2: political, medium, include, digital, australia, use, campaign, ai, disinformation, platform
Topic 3: vote, volunteer, poll, voter, booth, candidate, party, day, ppvc, pre
Topic 4: party, political, public, australia, democracy, donation, year, parliament, reform, independent
Topic 5: candidate, campaign, party, term, poll, would, behaviour, voter, political, number
dominant_topic
1     35
2     27
3    152
4     43
5     37
Name: count, dtype: int64
```

(Results of LDA-Topic Modelling)



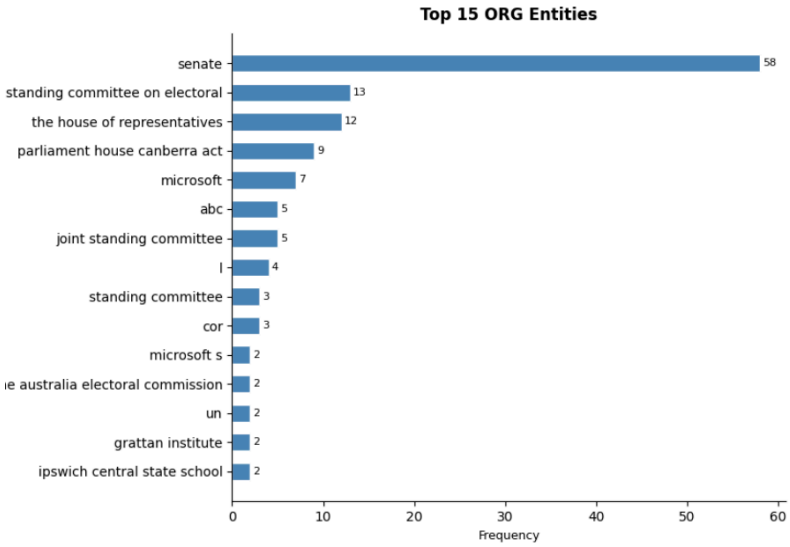
Insights: The five LDA topics paint a clear picture of what the 2025 federal election inquiry is really about. Topic 3 dominates overwhelmingly with 152 submissions its keywords volunteer, poll, booth, voter, ppvc, and pre confirm it captures concerns about the physical conduct of election day, directly answering **RQ1** that polling operations are the single most widespread public concern in this inquiry. Topic 2 is the most analytically significant despite having only 27 submissions. Its keywords disinformation, platform, digital, ai, and medium form a coherent cluster around online political communication and emerging technology risks, providing direct evidence for **RQ2** that concern about disinformation represents a distinct and identifiable strand of public opinion rather than scattered individual mentions. Topic 4 addresses structural democratic reform democracy, donation, parliament, reform, and independent

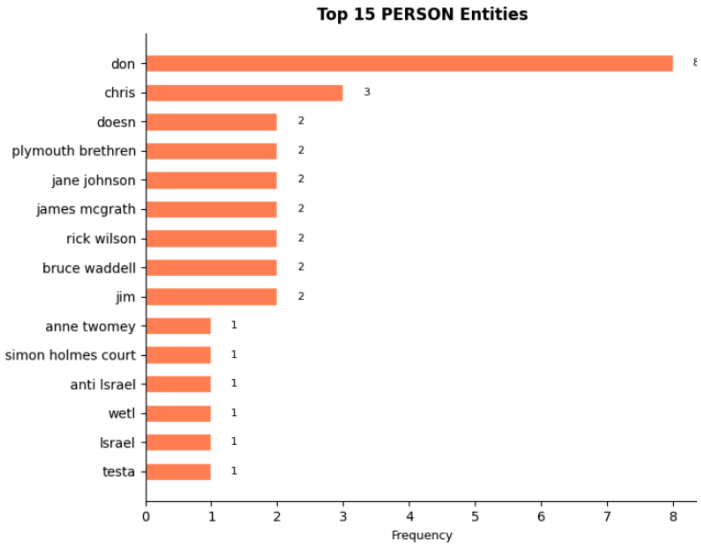
suggesting a group of submitters concerned less with how the election was run and more with the broader health of Australian democracy. Topic 1 covers senate and parliamentary representation while Topic 5 captures candidate conduct and campaign behaviour, together completing a picture of a corpus that is simultaneously concerned with electoral mechanics, institutional integrity, and democratic reform.

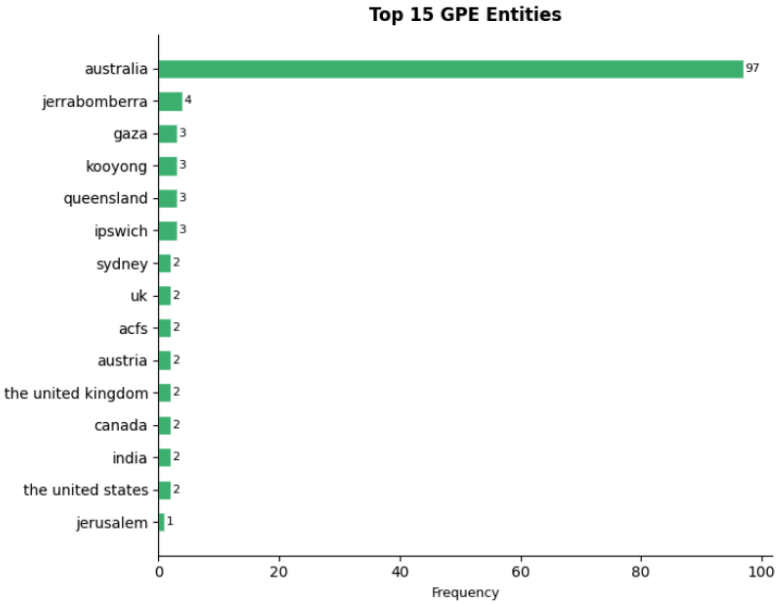
Section 8: Named Entity Recognition (NER) and Sentiment Analysis

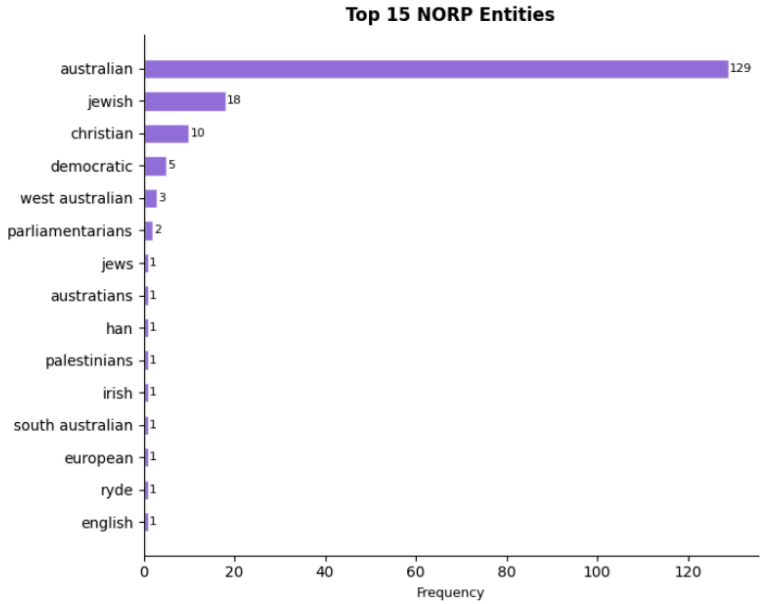
Named Entity Recognition (NER) uses spaCy's language model to automatically identify and classify real-world entities mentioned across the submissions including organisations, people, places, and nationalities. Rather than looking at what topics are discussed, NER reveals who and what submitters are specifically referencing, adding a layer of analysis that topic modelling cannot provide. Four entity types were selected for deeper analysis ORG(organizations), PERSON (people), GPE (Geopolitical Entity) , and NORP (National and Religious Entites) as these are the most analytically relevant for understanding the landscape of voices and references within an electoral inquiry.

Insights: Microsoft and ABC appearing in the top organisations connects to RQ2, likely referenced in the context of digital voting technology and political advertising respectively. The NORP results are the most significant for RQ3 Jewish (18) and Christian (10) communities ranking as the top group mentions suggests organised religious groups are disproportionately active in this inquiry, raising questions about whether the committee is hearing a truly representative cross-section of public opinion. This pattern reinforces a broader concern running through RQ3 institutions and organised groups, whether the AEC submitting 10,000 words or religious communities mobilising their members to submit, have a structural advantage over individual citizens writing alone. The GPE chart adds further context Gaza and Kooyong appearing alongside Australia signals that some submissions were motivated by very specific geopolitical and electorate-level concerns unique to the 2025 campaign.



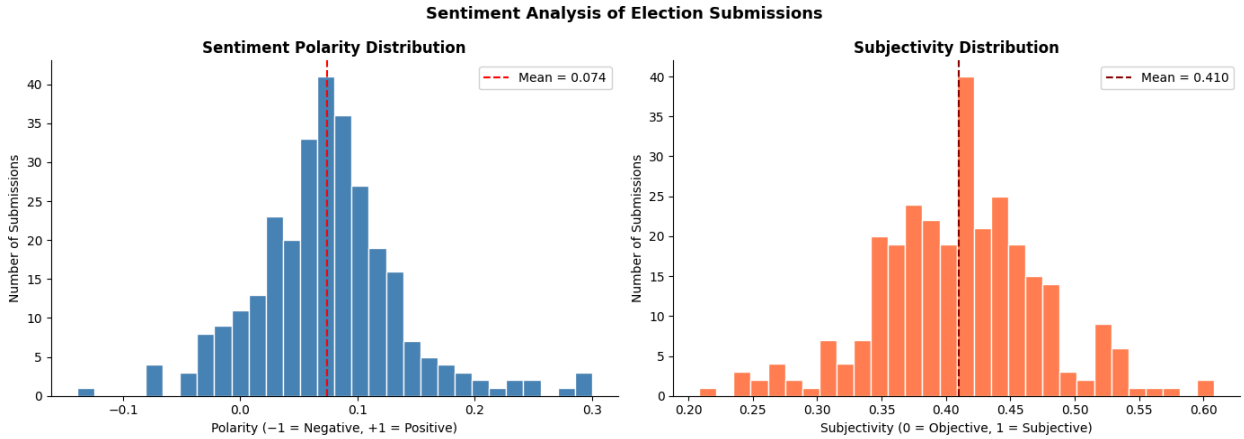






Sentiment analysis was conducted using TextBlob, which assigns each submission two scores polarity (ranging from −1 for very negative to +1 for very positive) and subjectivity (ranging from 0 for purely objective to 1 for highly subjective). This allows us to measure not just what people are talking about but how they feel about it, and whether their tone differs across the topics identified in the LDA analysis.

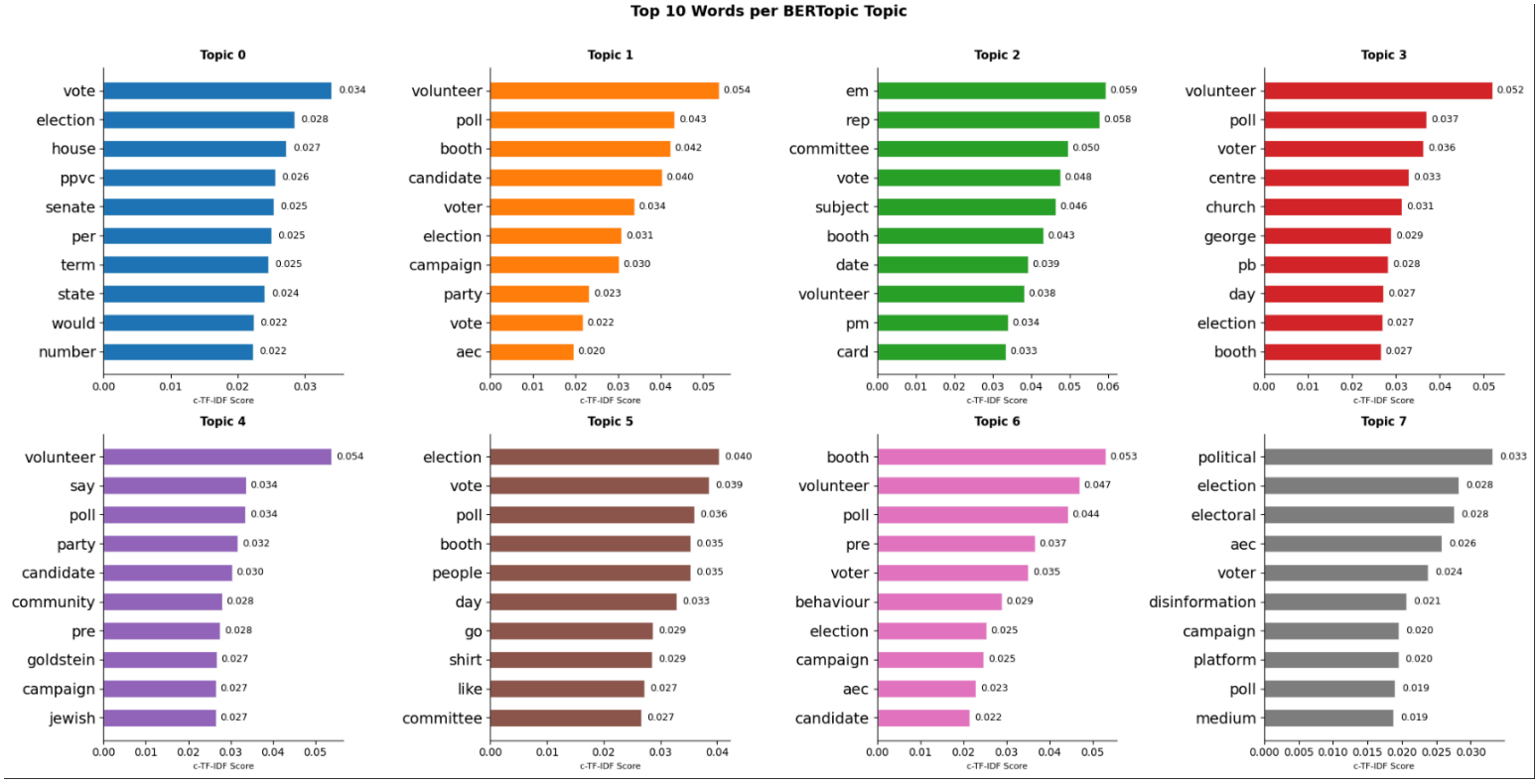
Insights: The mean polarity of 0.074 is surprisingly measured for a political dataset submissions are not angry, they are constructive, with most clustering tightly between 0.0 and 0.15. The subjectivity mean of 0.410 confirms people are sharing opinions rather than purely factual reporting, which is expected for a public consultation. Breaking sentiment down by LDA topic reveals that Topic 3 polling day operations, the largest topic has the lowest polarity (0.068), meaning the most widely shared concern is also expressed the most calmly. Topic 1, covering senate and parliamentary representation, scores highest (0.092), suggesting slightly more optimism among submitters focused on structural reform. Across all five topics sentiment remains consistent.



Section 9: BERTopic Modelling:

BERTopic uses sentence-transformer embeddings to represent each submission as a dense semantic vector, capturing the full context and meaning of text rather than just counting words. It then automatically clusters semantically similar submissions using HDBSCAN, meaning the number of topics does not need to be specified in advance the algorithm decides based on the data. This makes it a more flexible and arguably more honest approach than LDA for an unstructured corpus like this one.

Insights: BERTopic identified 8 distinct topics across the corpus, with 80 submissions classified as outliers (Topic -1) meaning roughly a quarter of submissions are too unique or short to cluster meaningfully, which itself says something about the diversity of individual voices in this inquiry relevant to **RQ3**. Topic 0 is the largest with 94 submissions and centres on general electoral and party politics, broader in scope than any single LDA topic. Topics 1, 3, 4, 5, and 7 all cluster around volunteering, polling booths, and pre-poll operations, which is striking. BERTopic has independently fragmented what LDA grouped as a single Topic 3 into multiple distinct sub-clusters, suggesting that polling day operations is not one homogeneous concern but several related but distinct issues being raised by different groups of submitters. This directly reinforces **RQ1**. Topic 2 stands out with its *em*, *rep*, and *committee* keywords, these are likely email formatting artefacts from submissions forwarded via email, pointing to a data quality issue worth acknowledging. Topic 5 introducing *Goldstein* as a keyword is the most specific finding Goldstein is a Melbourne electorate that became nationally prominent through the teal independent movement, suggesting a cluster of submissions specifically motivated by that local contest rather than general electoral reform, again connecting to **RQ3** and the question of organised community influence on the inquiry.



Section 10: Inference and Recommendations:

10.1 Inference:

RQ1: What are the dominant themes and concerns raised by the Australian public about the conduct of the 2025 federal election?

RQ2: Is there evidence of growing public concern about disinformation and online political advertising in Australian elections?

RQ3: Are all voices represented equally in this inquiry, or do institutional submitters dominate the conversation over individual citizens?

This project set out to answer three research questions about the 299 public submissions lodged with the 2025 Australian Federal Election inquiry, and the NLP pipeline applied across eight analytical methods consistently pointed toward the same answers.

On **RQ1**, the evidence is unambiguous. Polling day operations, pre-poll voting centres, booth management, volunteer conduct, and ballot procedures dominate the corpus at every level of analysis. From the raw word frequency counts to TF-IDF, from LDA Topic 3 capturing 152 submissions to BERTopic independently fragmenting that same concern into multiple distinct sub-clusters, every method confirmed that the physical mechanics of how Australians cast their votes is the single most widespread public concern raised in this inquiry. Reform appears in 115 submissions nearly half the working corpus confirming that this concern is not passive. People are not just describing problems, they are demanding change.

On **RQ2**, the answer is yes, but it is a qualified yes. LDA Topic 2, despite capturing only 27 submissions, produced a coherent and distinct cluster of keywords disinformation, platform, digital, ai, and medium that clearly represents an emerging strand of public concern about online political communication. The trigram truth in political advertising appearing 143 times reinforces this. Disinformation as a concern is real and organised, but it remains a minority voice within the corpus. It is an emerging issue rather than a dominant one which is itself an important finding, suggesting the regulatory and policy conversation around online political advertising is still in its early stages in Australia.

On **RQ3**, the analysis reveals a more nuanced picture than a simple yes or no. Individual citizens make up the majority of submissions by volume, but the distribution is deeply unequal in ways that matter. The AEC submitted 10,131 words against a median individual submission of 831. Religious organisations Jewish and Christian communities in particular appear disproportionately in the NORP entity analysis. BERTopic identified Goldstein as a distinct topic keyword, pointing to a cluster of submissions mobilised around a specific local electorate contest rather than general electoral reform. Eighty submissions were classified as outliers by BERTopic,

meaning a significant portion of individual voices are too unique to cluster; they are present in the corpus but effectively invisible in aggregate analysis. The inquiry formally welcomes all voices equally, but the data suggests some voices arrive with considerably more structural support than others.

10.2: Recommendations:

Three practical recommendations emerge from the analysis of the 2025 federal election inquiry submissions.

First, electoral administration reform should prioritise pre-poll voting operations, polling place management, and volunteer conduct. Across the project, this concern appeared with the greatest consistency and breadth. It was visible in the most frequent words, strongly represented in TF-IDF, repeated in the bigram and trigram analysis, and emerged as the dominant theme in both LDA and BERTopic. This convergence across methods suggests that operational issues at polling booths and pre-poll voting centres were not isolated complaints, but the most widely shared concern in the inquiry. Reform in this area should therefore be treated as the clearest practical priority arising from the submissions.

Second, disinformation and online political advertising should be recognised as a distinct electoral integrity issue rather than being absorbed into general reform debates. The analysis identified this concern not simply as scattered keywords, but as a coherent thematic cluster. In particular, the appearance of a separate disinformation-related topic in LDA, together with the repeated prominence of phrases such as truth in political advertising, indicates that submitters are increasingly framing digital misinformation and online campaigning as specific problems requiring direct policy attention. This suggests that future electoral reform discussions should address the online information environment as a standalone regulatory challenge.

Third, the inquiry process should account more explicitly for imbalance in participation between institutional and individual submitters. The findings show that while individual citizens contributed many submissions, institutional actors often submitted far longer, more detailed, and more formal documents, giving them greater visibility within the corpus. Named entity patterns also suggest that organised groups were particularly active in shaping the discussion. This does not mean institutional submissions are less valuable, but it does raise questions about whether all voices are heard on equal terms. Future inquiries could respond to this imbalance by reporting findings separately by submitter type, presenting citizen and institutional perspectives in parallel, and ensuring that repeated concerns raised by many individuals are not overshadowed by a smaller number of lengthy institutional documents.